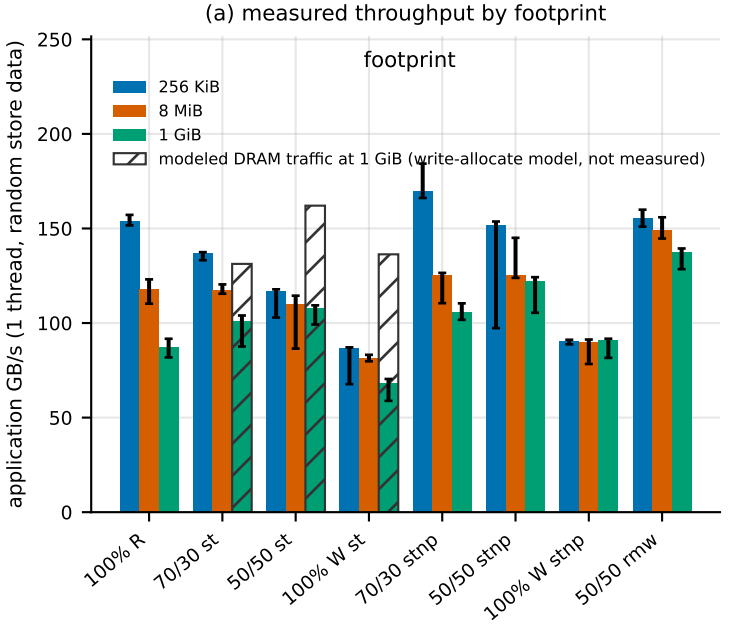
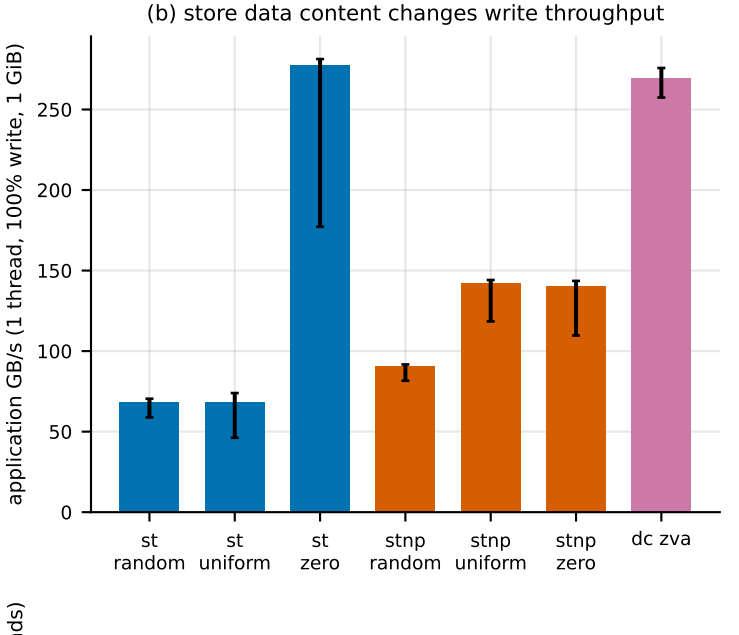
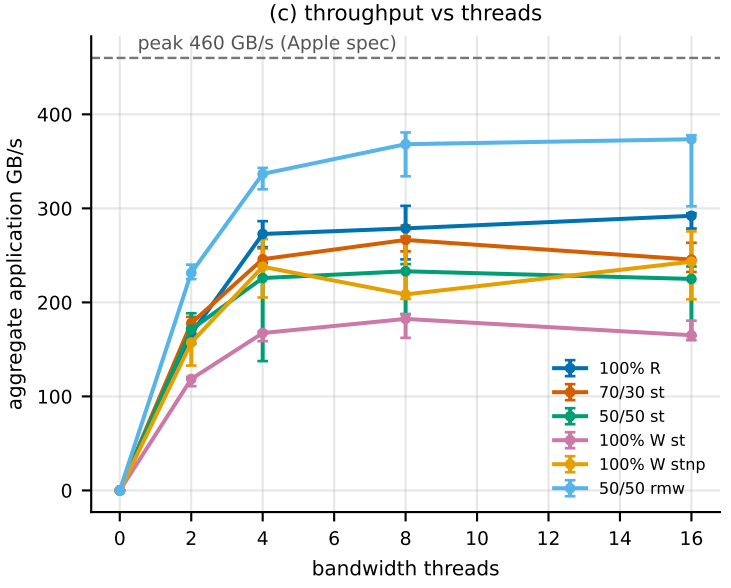
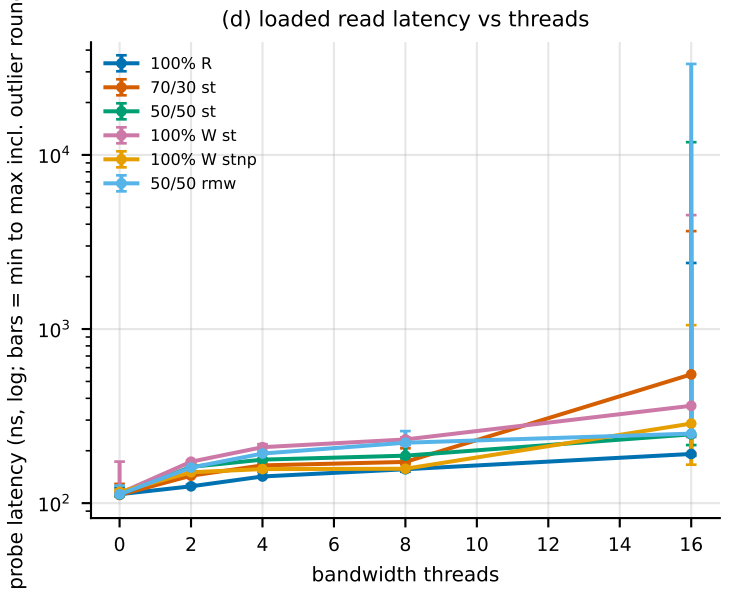

(a) measured throughput by footprint
footprint
256 KiB
8 MiB
1 GiB
modeled DRAM traffic at 1 GiB (write-allocate model, not measured)
application GB/s (1 thread, random store data)
0
50
100
150
200
250
100% R
70/30 st
50/50 st
100% W st
70/30 stnp
50/50 stnp
100% W stnp
50/50 rmw


(b) store data content changes write throughput
application GB/s (1 thread, 100% write, 1 GiB)
0
50
100
150
200
250
st random
st uniform
st zero
stnp random
stnp uniform
stnp zero
dc zva


(c) throughput vs threads
peak 460 GB/s (Apple spec)
aggregate application GB/s
0
100
200
300
400
bandwidth threads
0
2
4
6
8
10
12
14
16
100% R
70/30 st
50/50 st
100% W st
100% W stnp
50/50 rmw


(d) loaded read latency vs threads
probe latency (ns, log; bars = min to max incl. outlier rounds)
10^2
10^3
10^4
bandwidth threads
0
2
4
6
8
10
12
14
16
100% R
70/30 st
50/50 st
100% W st
100% W stnp
50/50 rmw